

Eight arms, three intervention sites

Within each task: the same frozen checkpoint and H3 hook timing; native has no edit.

Arm	R · B3 output	V · visual input	A · B3 condition
Native	—	—	—
Refined four-direction	learned	—	—
Calibrated random subspace	random	—	—
Equal-budget coupling	—	scaled learned	scaled learned
Dose-matched random directions	—	scaled random	scaled random
Unscaled joint	—	original dose	original dose
Visual only	—	original dose	—
Action-conditioning only	—	—	original dose

Scaled: each V/A component uses $1/\sqrt{2}$ of its original standardized dose. R arms never include V or A.
Compare learned R with its calibrated random subspace; compare learned V/A with dose-matched random directions.